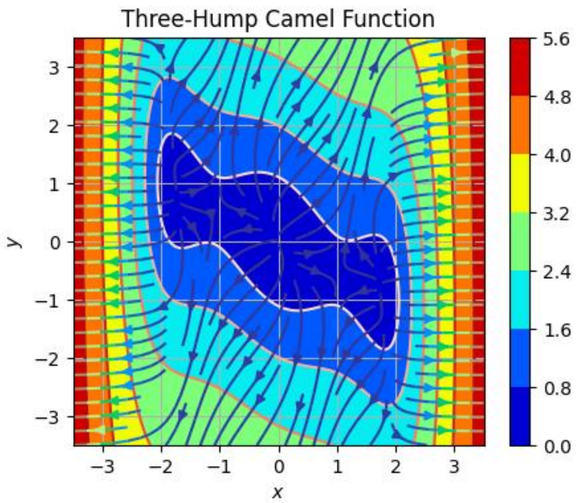
# 112A Introduction to Computer and Computer Science

## Homework Assignment #9

**Due: 12/15 12:00:00**

In this homework assignment, you will practice how to apply the concept of object-oriented programming to solve a practical issue. As you have learned all the related knowledge during the lectures, you should have the ability to accomplish this.

**Problem #1: Three-Hump Camel Function (50%)**



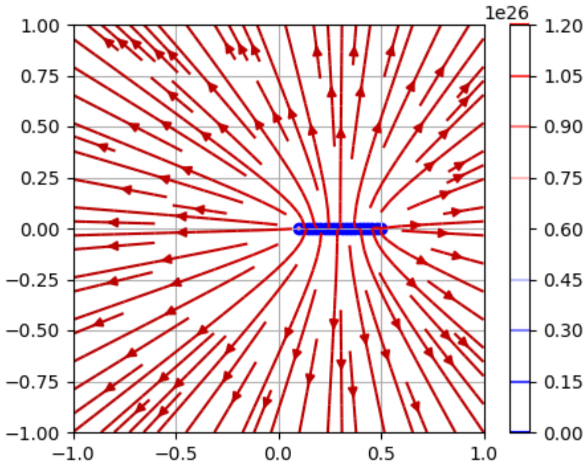

Please write a program to plot the contour map and draw the streamlines of its vector flow for the Three-Hump Camel function:

$$f(x,y) = 2x^2 - 1.05x^4 + \frac{x^6}{6} + xy + y^2$$

Please notice that,

1. The range of your plot should be within $[-3.5, 3.5]$ for both axes.
2. You don't need to follow the style of this example.

**Problem #2: Sticks with $N$ positive charges (50%)**




Please write a program to plot the contour map and draw the streamlines of its vector flow for $N$ positive charges at $x = [0.1, 0.5]$. Again, you don't need to follow the style of this example.

Please accomplish this homework with an organized code (e.g., with main script and function script). **Your main script file name must be 'main_hw8.py.'** Here is a template for your code structure:

```
112A_hw9_0123456789
      ├─ obj.py          # Module 1 for hw9
      ├─ func.py         # Module 2 for hw9
      ├─  :              # Other modules if necessary
      └─ main_hw8.py     # Main script of hw9
```

**FRIENDLY REMINDER**

You don't need to follow this structure exactly, just keep your main script clean.

Here is a template for your main script.

```python
import argparse
from func import solution1, solution2


if __name__ == "__main__":
    parser = argparse.ArgumentParser(description="2023 NYCUDOPCS
Homework #8")
    parser.add_argument("--save", default=False, type=bool, help="Save
figure? (default: False)")

    opt = parser.parse_args()


    print("Problem 1:")
    solution1(saveFigure=opt.save)
    print("="*60)


    print("Problem 2")
    solution2(saveFigure=opt.save)
    print("="*60)
```

**Hand in procedure:**

As we had mentioned in the lecture, you should list all your collaborators in your programs. Here is the template:

```
"""
Created on Sun Aug 7 01:23:45 2023

@author: Xi Winnie, student ID

@collaborators: EE Tsai, her student ID
"""
```

Please save your homework as a "**.zip**", "**.7z**", or "**.rar**" file, where the file name should follow this format:

112A_hw9_ID.zip

For example,

112A_hw9_0123456789.zip

Please note. **We are not going to accept any homework file with wrong file name, wrong file format, or without signature in main script**. Please double check the content of your file.

Once you have accomplished your works, you can submit your homework to the "E3@NYCU" system. There will be a section for uploading your homework.